Math 105L: Using First and Second Derivatives II

Lecture 11-1 Completed Board Lecture & Discussion Sheet

Duke University | Department of Mathematics | Monday, November 2, 2026

1. Theoretical Foundation: Extrema vs. Inflection Points

First Derivative Test vs. Points of Inflection

- **Local Extrema (First Derivative Test)**: Let a be in the interior of the domain of continuous f. If $f'(x)$ changes sign across a, then $(a, f(a))$ is a local extremum. Specifically, a transition from positive to negative $(+ \to -)$ yields a **local maximum**, while negative to positive $(- \to +)$ yields a **local minimum**. A critical point ($f' = 0$ or undefined) is a *candidate*, but not sufficient on its own.
- **Points of Inflection**: A point $(c, f(c))$ on the curve is an **inflection point** if f is continuous at c and the **concavity changes sign** across c (f'' transitions $+ \to -$ or $- \to +$). The algebraic condition $f''(c) = 0$ (or undefined) identifies *candidates*, but concavity must strictly alternate across the point.

Common Exam Fallacy: Setting $f''(x) = 0$ Does NOT Find Extrema!

Setting $f''(x) = 0$ identifies candidate inflection points, **not** local extrema. Local extrema occur exclusively at critical points ($f' = 0$ or undefined) where f' changes sign. The **Second Derivative Test** evaluates the sign of $f''(a)$ at an *already known* critical point where $f'(a) = 0$ ($f''(a) > 0 \implies$ local min; $f''(a) < 0 \implies$ local max; $f''(a) = 0 \implies$ inconclusive).

Comparative Benchmark: Why Candidates Must Be Tested for Sign Changes

Function	$f'(x)$	$f''(x)$	At $x = 0$	Geometric Classification at $x = 0$
$y = x^3$	$3x^2 \geq 0$	$6x$	$f'(0) = 0, f''(0) = 0$	**Inflection Point** (f'' changes sign $- \to +$). Neither local max nor min.
$y = x^4$	$4x^3$	$12x^2 \geq 0$	$f'(0) = 0, f''(0) = 0$	**Local Minimum** (f' changes $- \to +$). *NOT* an inflection point ($f'' \geq 0$).
$y = -x^4$	$-4x^3$	$-12x^2 \leq 0$	$f'(0) = 0, f''(0) = 0$	**Local Maximum** (f' changes $+ \to -$). *NOT* an inflection point ($f'' \leq 0$).

2. Case Study: High-Degree Polynomial $f(x) = (x^2 - 4)^7$

Global Algebraic Structure:

- **Degree & Parity**: Degree $2 \times 7 = 14$ (even degree, positive leading coefficient $\implies \lim_{x \to \pm\infty} f(x) = +\infty$). Even function ($f(-x) = f(x)$), reflecting bilateral symmetry across the y-axis.
- **Roots (x-intercepts)**: Setting $f(x) = 0 \implies (x^2 - 4)^7 = 0 \implies x^2 - 4 = 0 \implies \mathbf{x = \pm 2}$.

First Derivative & Critical Point Classification: Applying the Generalized Power Rule (Chain Rule):

$$f'(x) = 7(x^2 - 4)^6 \cdot \frac{d}{dx}[x^2 - 4] = 7(x^2 - 4)^6 \cdot (2x) = \mathbf{14x(x^2 - 4)^6}$$

Setting $f'(x) = 0$ yields critical points at $14x = 0 \implies \mathbf{x = 0}$ or $(x^2 - 4)^6 = 0 \implies \mathbf{x = \pm 2}$.

First Derivative Sign Mechanics: Because $(x^2 - 4)^6$ has an even exponent (6), it is **strictly non-negative** for all real numbers ($(x^2 - 4)^6 \geq 0$). Consequently, the sign of $f'(x)$ is governed entirely by the linear factor $14x$:

- On $(-\infty, 0)$: $x < 0 \implies f'(x) < 0$, so $f(x)$ is **strictly decreasing**.
- On $(0, \infty)$: $x > 0 \implies f'(x) > 0$, so $f(x)$ is **strictly increasing**.
- **At** $x = 0$: f' transitions from negative to positive $(- \to +) \implies$ **Strict Local Minimum** at $(0, (-4)^7) = (0, -16384)$.
- **At** $x = \pm 2$: f' attains zero but does **not** change sign ($f' \leq 0$ on $(-\infty, 0]$ and $f' \geq 0$ on $[0, \infty)$) $\implies$ **Horizontal shelf points (Neither local max nor min)!**

2. Case Study: High-Degree Polynomial (Continued)

Second Derivative Derivation via Product & Chain Rules: Differentiating $f'(x) = 14x \cdot (x^2-4)^6$:

$$\begin{aligned} f''(x) &= \frac{d}{dx}[14x] \cdot (x^2-4)^6 + 14x \cdot \frac{d}{dx}\left[(x^2-4)^6\right] \\ &= 14(x^2-4)^6 + 14x \cdot \left[6(x^2-4)^5 \cdot (2x)\right] = 14(x^2-4)^6 + 168x^2(x^2-4)^5 \\ &= 14(x^2-4)^5\left[(x^2-4) + 12x^2\right] = \mathbf{14(x^2-4)^5(13x^2-4)} \end{aligned}$$

The Factoring Rule: Always factor out the greatest common factor $14(x^2-4)^5$. Expanding $(x^2-4)^6$ would lead to an unmanageable degree-13 polynomial!

Setting $f''(x) = 0$ yields two sets of candidate inflection boundaries:

$$(x^2-4)^5 = 0 \implies \mathbf{x = \pm 2}, \qquad \text{and} \qquad 13x^2 - 4 = 0 \implies x^2 = \frac{4}{13} \implies \mathbf{x = \pm\frac{2}{\sqrt{13}}}.$$

Why $\frac{2}{\sqrt{13}} < 2$ (Strictly Without Calculators): To order our candidates along the real line without decimal approximations, observe that:

$$13 > 4 \implies \sqrt{13} > \sqrt{4} = 2 \implies \frac{2}{\sqrt{13}} < \frac{2}{2} = 1 < 2.$$

Because $\sqrt{13} > 2$, dividing 2 by a number strictly greater than 2 yields a value strictly less than 1, which is strictly less than 2. By bilateral symmetry, $-\frac{2}{\sqrt{13}}$ lies strictly between -2 and 0. Thus, the candidate roots are strictly ordered along the real line as:

$$\mathbf{-2 < -\frac{2}{\sqrt{13}} < \frac{2}{\sqrt{13}} < 2}.$$

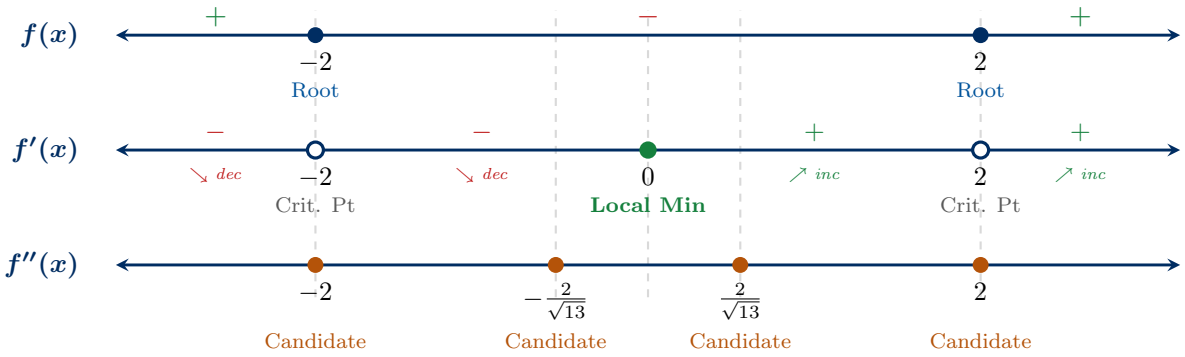
The Non-Calculator Global View: Signs via Parabola Geometry

The Primary Course Objective: In Math 105L, our goal is to execute complete curve sketching and concavity analysis **entirely without calculators**. Rather than testing arbitrary test points, we adopt a **global geometric view** by observing the shape and orientation of each factor across the whole real line:

- **Factored Second Derivative**: $f''(x) = 14 \cdot (x^2-4)^5 \cdot (13x^2-4)$ is the product of a positive scalar ($14 > 0$) and **two quadratic expressions**:
  1. **Factor 1**: $(x^2-4)^5$. The base is the parabola $y = x^2 - 4$. Because its leading coefficient is positive ($+1 > 0$), it is an **upward-opening parabola (tails pointing up)** with roots at $x = \pm 2$. Geometrically, any upward-opening parabola is **positive outside its roots** ($|x| > 2$) and **negative between its roots** ($-2 < x < 2$). Because 5 is an odd exponent, $(x^2-4)^5$ preserves this exact sign signature everywhere!
  2. **Factor 2**: $(13x^2-4)$. This is also an **upward-opening parabola (tails pointing up)**, having leading coefficient $13 > 0$ and roots at $x = \pm\frac{2}{\sqrt{13}}$. By identical parabola geometry, it is **positive outside its roots** ($|x| > \frac{2}{\sqrt{13}}$) and **negative between its roots** ($|x| < \frac{2}{\sqrt{13}}$).
- **The Global Multiplying Rule**: Because **both parabolas have their tails pointing up**, both factors are positive as $x \to \pm\infty$ and dip negative precisely between their roots. We determine the sign of $f''(x)$ across all five intervals by pure geometry with **zero calculator arithmetic**!

3. Aligned Comparative Number Lines: $f(x)$, $f'(x)$, and $f''(x)$

The diagram below connects key x-coordinates vertically across all three functions. All points are rendered as completely round circular markers, and dashed cords illustrate the geometric alignment.



Note: Candidate boundaries established from $f''(x) = 0$. Interval sign testing across the 5 open intervals will be completed next lecture using Parabola Geometry.

4. Prerequisite Prior Worksheets & Resource Portals

Connected Course Materials & Submission Portals

To solidify the algebraic foundations, leading behavior, and curve analysis used today, review these prior worksheets:

- **Worksheet 2-1: Power Functions, Monomials, & Polynomials (Week 2)**
  - **Why It Connects**: Dominant terms as $x \to \pm\infty$, even vs. odd polynomial parity, root multiplicity, and sign preservation in power factors (e.g., $(x^2-4)^6 \geq 0$ vs. $(x^2-4)^5$).
  - **Courseware Links**: **Completed Summary (PDF)** | **Panopto Video ↗** | **Original Worksheet (PDF)** | **Gradescope Sub 1** | **Gradescope Resub**
- **Worksheet 10-3: Using First and Second Derivatives I (Week 10)**
  - **Why It Connects**: Establishing critical points, classifying extrema via the First Derivative Test, and identifying candidates for inflection points.
  - **Courseware Links**: **Completed Summary (PDF)** | **Panopto Video ↗** | **Original Worksheet (PDF)** | **Gradescope Sub 1** | **Gradescope Resub**

5. Key Conceptual Takeaways & Homework Assignment

Executive Takeaways & Practice for Homework

- **Critical Points vs. Extrema**: $f'(x) = 14x(x^2-4)^6$ has three critical points ($x = 0, \pm 2$). Only $x = 0$ is a true extremum (local minimum) because f' changes sign. At $x = \pm 2$, f' does not change sign, producing horizontal shelf/terrace points.
- **The Factoring Rule**: When computing higher derivatives of composite functions, always factor out the lowest power of the repeated factor $(14(x^2-4)^5)$. Never expand!
- **Candidates for Inflection**: Setting $f''(x) = 0$ produces four candidates: $x = \pm 2$ and $x = \pm\frac{2}{\sqrt{13}}$.
- **Homework Next Steps**: For homework, apply the Parabola Geometry rule across the five open intervals:

$$(-\infty, -2), \quad \left(-2, -\frac{2}{\sqrt{13}}\right), \quad \left(-\frac{2}{\sqrt{13}}, \frac{2}{\sqrt{13}}\right), \quad \left(\frac{2}{\sqrt{13}}, 2\right), \quad (2, \infty)$$

to prove that concavity alternates across all four candidates and complete the global curve sketch of $f(x)$.